COMPUTER AIDED ENGINEERING DRAWING

For the Academic year 2021 – 2022

DEPARTMENT OF MECHANICAL ENGINEERING

QUESTION BANK

RAMAIAH INSTITUTE OF TECHNOLOGY

(Autonomous Institute, Affiliated to VTU)

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CHAPTER 1. PROJECTION OF POINTS

1. A point P is 30mm in front of VP, 40mm above HP and 50mm From RPP Draw its projections.
2. The point P is 45mm above HP, 60mm infront of VP and 30mm from RPP Draw the three principle view of the point also state the quadrant in which it lies.
3. A Point a is 20mm above HP and in the first quadrant its shortest distance from the XY line is 40mm. Draw the projections. Determine the distance form VP.
4. A Point P is on HP and 30mm in front of VP. Another Point Q is on VP and 40mm above HP The distance between their projections parallel to XY line is 50mm. Find the distance between their front and top views of the points P & Q.
5. A Point is 35mm below HP, 15mm behind VP & 25mm behind RPP. Draw its projections
6. Draw all the three views of a point P lying 40mm on HP, 30mm in front of VP and 20mm from the RPP. Also state the quadrant in which it lies.
7. A Point p is 25mm above HP & 20mm in front of VP. Another point Q is on HP and 30mm behind VP. The distance between their projections measured parallel to the line of intersection of VP and HP is 50mm. Find its distance between the top views of points P and Q.
8. A point P is on HP and 35mm infront of VP. Another Point Q is on VP and below HP. The line joining their front views makes an angle of 30deg to XY line, while the line joining their top views makes an angle of 45^0 with XY line. Find the distance of the point Q from HP.
9. Two points R and S are on HP. The point R is 35mm infront of VP, while S is 50mm behind VP. The line joining their top views makes an angle of 40^0 with XY. Find the horizontal distance between the two projectors.
10. Two points P and Q are on HP. The point P is 30mm behind VP, while Q is 50mm infront of VP. The line joining their top views makes an angle of 40^0 with XY. Find the horizontal distance between the two projectors.

CHAPTER 2. PROJECTION OF LINES

1. A Line AB 80mm long has its end A 20mm above the HP and 30mm infront of VP. It is inclined at 30° to HP and 45° to VP Draw the projections of the line and find apparent lengths and apparent inclinations.
2. A line 80mm long has one of its ends both in HP and VP, and inclined at 30° to HP and 45° to VP. Draw its Projections.
3. A line PQ 75mm long has one end in VP and the other end Q in HP. The line is inclined at 30° to HP and 60° to VP. Draw its projections.
4. A Line AB has its end A 20mm above the HP and 30mm infront of VP. The other end B is 60mm above the HP. The distance between the end projectors is 70mm. Draw its projections. Determine the apparent inclination.
5. The top view of a line PQ,75mm long measures 50mm. The end P is 30mm ind\front of VP and 15mm above HP. The end Q is 15mm infront of VP and above HP. Draw the projections of the line PQ and find its true inclination with HP and VP.
6. A Line PQ 85mm long has its end P 10mm above the HP and 15mm infront of the VP. The top view and front view of line PQ are 75mm and 80mm respectively. Draw its projections. Also determine the true and apparent inclination of the line.
7. A line has its end A 10mm above HP and infront of VP. The end B is 55 mm above HP and line is inclined at 30° to HP and 35° to VP. The distance between the end projectors is 50mm. Draw the projections of the line. Determine the true length of the line and its inclination with VP.
8. The front view of a 90mm long line which is inclined at 45° to the XY line measures 65mm. End A is 15mm above the XY line and is in VP. Draw the projections of the line and find its inclination with HP and VP
9. A line AB 60mm long has one of its extremeties 20mm infront of VP and 15mm above HP. The line is inclined at 25° to HP and 40° to VP. Draw its top and front views.
10. A Line AB measuring 70mm has its end A 15mm infront of VP and 20mm above HP and the other end B is 60mm infront of VP and 50mm above HP. Draw the projection of the line an find the inclinations of the line with both the reference planes of projections.

CHAPTER 3. PROJECTION OF PLANE SURFACES

1. An equilateral triangular lamina of 30mm sides lies with one of its edges on HP such that the surface of the lamina is inclined to HP at 60^o. The edge on which it rests is inclined to VP at 60^o. Draw the projections.

2. ABC is an equilateral triangular plate of 40mm sides. It is lying on its edge BC on HP which is making an angle of 45^o with VP. The two edges AB and AC measures 30mm in the top view. Determine the slope angle of the plate with HP and draw its front view.

3. An isosceles triangular plate has base 50mm long and altitude 70mm. It is so placed that in the front view it is seen as an equilateral triangle of 50mm side. The resting side is inclined at 45^o to the HP. Draw its top and front views.

4. A mirror 60x80mm is inclined to the wall at such an angle that its front view is a square of 60mm side. The longer sides of the mirror appears perpendicular to both HP and VP. Find the inclination of the mirror with the wall.

5. A pentagonal lamina of 20mm sides is resting on HP with one of its corner touching it, such that the surface makes an angle of 60^o to HP. The two edges containing the corner on which it rests make equal inclinations with HP. When the edge opposite to this corner makes 45^o with VP and nearer to the observer, draw projections.

6. A pentagonal lamina of sides 25mm is having a side both on HP and VP. The corner opposite to the side on which it rests is 15mm above HP. Draw the top and front views of the lamina.

7. A hexagonal lamina of 30mm sides rests on HP with one of its corners on HP and surface inclined at 45^o to HP. The top view of the diagonal appears to be inclined at 30^o to VP . Draw the front and top views of the lamina.

8. A regular hexagonal lamina of sides 30mm is lying in such a way that one of its sides touches both the reference planes. If the side opposite to the side on which it rests is 45mm above HP, draw the projections of the lamina.

9. A hexagonal lamina of sides 25mm rests on one of its corner on HP. The corner opposite to the corner on which it rests is 35mm above HP and the diagonal passing through the corner on which it rests is inclined at 30^o to VP. Draw its projections. Find the inclination of the surface with HP.

10. Draw the projections of a circular plate of negligible thickness of 50mm diameter resting on HP on a point on the circumference, with its plane inclined at 45^o to HP and the top view of the diameter passing through the resting point makes 60^o with VP.

CHAPTER 4. PROJECTIONS OF SOLIDS

1. A Square Prism of 35mm sides of base and 65mm axis length rests on HP. Draw the projections of the prism when the axis is inclined to HP at 45° and axis appears to be inclined at VP at 30°.
2. A Pentagonal prism 25mm sides of base and 60mm axis length rests on HP. The rectangular face is inclined at 45° to HP and on one of its edges of the base which is inclined to VP at 30°. Draw the projections of the prism.
3. A Hexagonal prism 25mm sides of base and 60mm axis length rests on HP on one of its edges of the base. Draw the projection of the prism when the axis is inclined to HP at 45° and appears to be inclined to VP at 40°.
4. A Pentagonal prism 25mm sides of base and 60mm axis length rests on HP with its corner. The longer edge of the prism is inclined at 45° to HP and on one of its edges of the base which is inclined to VP at 30°. Draw the projections of the prism.
5. A cylinder of 50mm diameter, 60mm height rests on HP in such a way that the axis is inclined to HP at 45° and appears to be inclined to VP at 40°.
6. A Square pyramid 35mm sides of base and 65mm axis length rests on HP on one of its edges of the base which is inclined to VP at 30°. Draw the projections of the pyramid when the axis is inclined to HP at 45°.
7. A Pentagonal pyramid 35mm sides of base and 65mm axis length rests on HP with on one of its edges. The triangular face of the pyramid is inclined to HP at 45° and the axis appears to be inclined to VP at 30°. Draw the projections of the pyramid .
8. A Square pyramid 35mm sides of base and 60mm axis length rests on HP on one of its corners of the base such that the two base edges containing the corner on which it rests makes equal inclination with HP. Draw the projections of the pyramid when the axis of the pyramid is inclined at 40° to HP and appears to be inclined to VP at 45°.
9. A Hexagonal pyramid 35mm sides of base and 60mm axis length rests on HP on one of its corners of the base such that the inclined edge is lying on HP. Draw the projections of the pyramid when the top view of an axis is inclined at to VP at 45°.
10. A Cone of 50mm base diameter and 60mm axis length rests on HP on one of its generators. Draw the projections when the axis appears to be inclined at 30° to VP.

CHAPTER 5. ISOMETRIC PROJECTION

1. Draw the isometric projection of a rectangular prism of 60x80x20mm thick surmounted a triangular pyramid of side 45mm wit a height of 50mm such that the axes of the solids are collinear.

2. A square pyramid of base side 40mm and height 70 mm rests symmetrically on a cube of edges 50mm. Draw the isometric projections of the combination of solids if the axes of the three solids are in common line.

3. On a square prism of sides 70mm and 25mm high, another square prism of 30mm sides and 40mm high is placed vertically and symmetrically. Draw the isometric drawing of the combination of the solids.

4. A cylinder slab 75mm diameter and 50mm thick is surmounted by a cube of 40mm side The axes of the solids are coaxial. Draw the isometric drawing.

5. A square pyramid with base side 85mm and height 125mm is resting on a cube of sides 100mm, the axes of both the cube and the pyramid are co axial two sides of the base of the pyramid are parallel to the edges of the cube. Draw the isometric drawing.

6. A cube of side 25mm is resting centrally on a rectangular slab 100mmx40mm and 30mm thick . Draw the isometric projections of the combinations.

7. Three cubes of sides 60mm. 40mm and 20mm are placed centrally one above the other in the ascending order of their side. Draw the isometric projection of the combination.

8. A cone of base diameter 50mm and height 60mm is placed centrally on the top face of a pentagonal prism side 45mm Draw the isometric projection of the combination.

9. A frustum of pentagonal pyramid of base 50mm, top 25mm and height 50mm is placed centrally on the top face of a cylinder 60mm and height 60mm. Draw the isometric projection of the combination.

10. A rectangular pyramid of base 40mmx25mm and height 50mm is placed centrally on a rectangular slab sides 100mmx60mm and thickness 20mm. Draw the isometric projection of the combination.